

The Program in Action

When you run *deepcave.py*, the output will look like this:

```
Deep Cave, by Al Sweigart al@inventwithpython.com
Press Ctrl-C to stop.
#####
#####
#####
#####
#####
#####
#####
#####
#####
#####
--snip--
```

How It Works

This program takes advantage of the fact that printing new lines eventually causes the previous lines to move up the screen. By printing a slightly different gap on each line, the program creates a scrolling animation that looks like the viewer is moving downward.

The number of hashtag characters on the left side is tracked by the `leftWidth` variable. The number of spaces in the middle is tracked by the `gapWidth` variable. The number of hashtag characters on the right side is calculated from `WIDTH - gapWidth - leftWidth`. This ensures that each line is always the same width.

```
1. """Deep Cave, by Al Sweigart al@inventwithpython.com
2. An animation of a deep cave that goes forever into the earth.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, beginner, scrolling, artistic"""
5.
6.
7. import random, sys, time
8.
9. # Set up the constants:
10. WIDTH = 70 # (!) Try changing this to 10 or 30.
11. PAUSE_AMOUNT = 0.05 # (!) Try changing this to 0 or 1.0.
12.
13. print('Deep Cave, by Al Sweigart al@inventwithpython.com')
14. print('Press Ctrl-C to stop.')
15. time.sleep(2)
16.
17. leftWidth = 20
18. gapWidth = 10
19.
20. while True:
21.     # Display the tunnel segment:
22.     rightWidth = WIDTH - gapWidth - leftWidth
23.     print(('#' * leftWidth) + (' ' * gapWidth) + ('#' * rightWidth))
```